Campus Café — Infrastructure architecture

Detail of the v1 data-flow design · same 3 functions, same flow · George Brown College · Work Integrated Project · October 2026

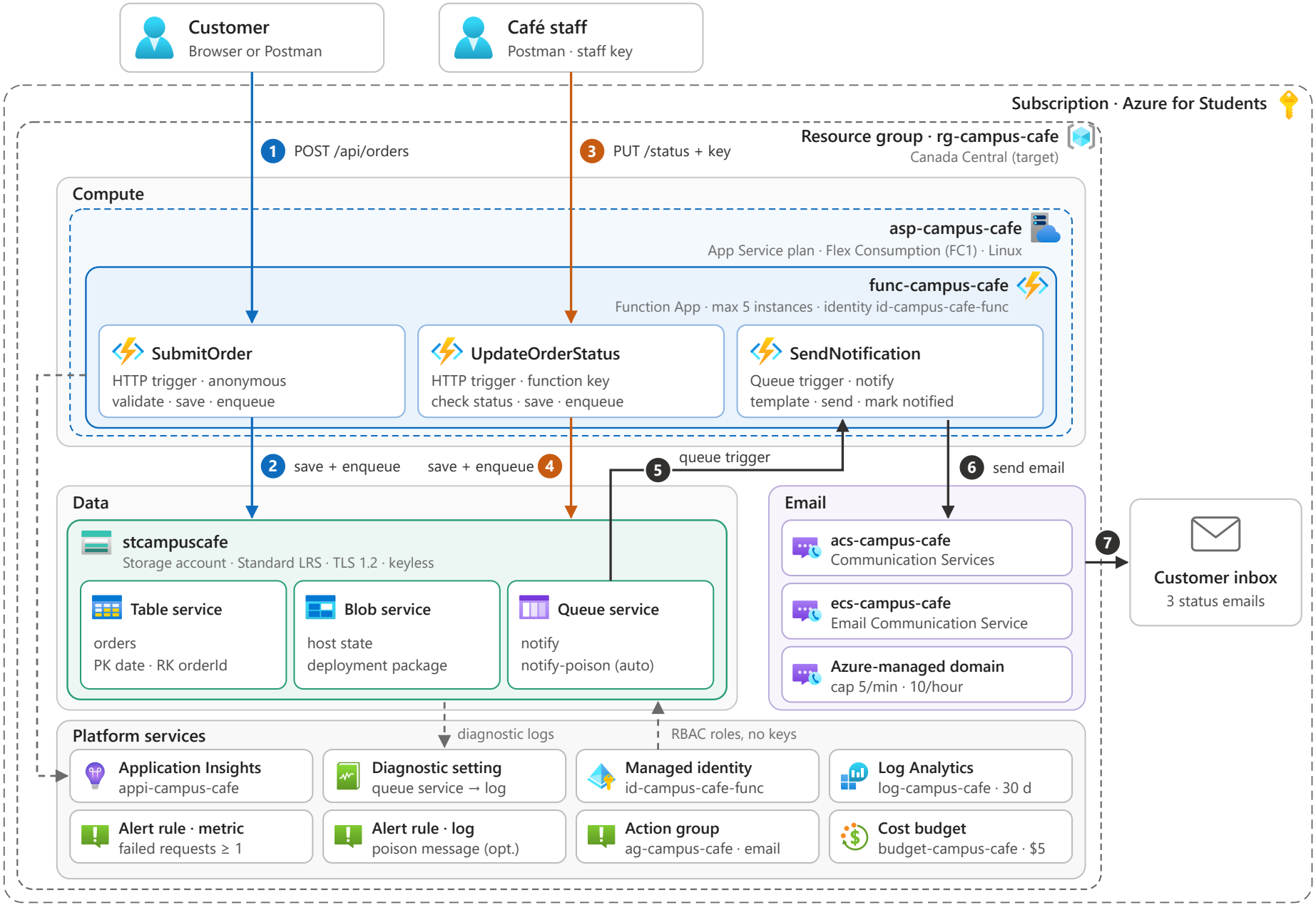


Figure 1. Subscription, resource group and every resource behind the v1 diagram. Numbers 1–7 match v1: customer order = 1–2 (blue), staff status update = 3–4 (orange), notification = 5–7 (shared). Dashed lines = identity (RBAC), diagnostic logs and telemetry. Azure icons: Microsoft Azure Public Service Icons V24, see References on page 3.

Resource inventory and access

18 parts behind the v1 diagram. "Implied" = required for a v1 box to work, not drawn in v1 itself.

#	Name	Azure resource type	Group	Key settings	In v1 diagram
1	`budget-campus-cafe`	Cost Management budget	Subscription	Email alert at $5 spend (support on Azure for Students unverified)	Yes
2	`rg-campus-cafe`	Resource group	—	Canada Central (target); tags app, env, course	Yes
3	`asp-campus-cafe`	App Service plan, Flex Consumption (FC1)	Compute	Linux; one app per plan	Implied
4	`func-campus-cafe`	Function App	Compute	3 functions; max 5 instances; always-ready 0; HTTPS only; CORS origins only if a browser page calls the API	Yes
5	`id-campus-cafe-func`	User-assigned managed identity	Platform	Attached to the Function App; RBAC on storage; created by the Flex portal wizard	Yes
6	`stcampuscafe`	Storage account	Data	Standard LRS; TLS 1.2; shared-key access off	Yes
7	table `orders`	Table service	Data	PK date, RK orderId; status, customer email, notified flags	Yes
8	queue `notify` + `notify-poison`	Queue service	Data	Poison after 5 tries (automatic); set `visibilityTimeout` 30–60 s (default 0 = instant retry)	Yes
9	blob containers	Blob service	Data	Functions host state + deployment package (Flex)	Yes
10	`acs-campus-cafe`	Communication Services	Email	Azure-managed domain linked	Yes
11	`ecs-campus-cafe`	Email Communication Service	Email	Holds the domain (no Cloud Adoption Framework prefix, team convention)	Implied
12	Azure-managed domain	Email domain (child resource)	Email	5 emails/min, 10/hour per subscription, no quota increase	Yes
13	`log-campus-cafe`	Log Analytics workspace	Platform	30 d retention; daily cap	Implied
14	`appi-campus-cafe`	Application Insights	Platform	Workspace-based; logs and failures from the Function App	Yes
15	diagnostic setting	On the storage queue service	Platform	StorageRead / Write / Delete → `log-campus-cafe`	Implied
16	alert rule · metric	Azure Monitor metric alert	Platform	Application Insights failed requests ≥ 1 in 5 min	Yes
17	alert rule · log (optional)	Azure Monitor log search alert	Platform	PutMessage to `notify-poison` in `StorageQueueLogs`; no free tier	Yes
18	`ag-campus-cafe`	Action group	Platform	Email to the team	Implied

Globally unique names (`func-` , `st…`) get a short suffix. Prefixes follow the Microsoft Cloud Adoption Framework abbreviations.

Access (RBAC, keyless)

Principal	Role	Scope	Why
`id-campus-cafe-func`	Storage Blob Data Owner	`stcampuscafe`	Functions host (documented minimum for the host storage connection) and deployment container
`id-campus-cafe-func`	Storage Table Data Contributor	`stcampuscafe`	Host diagnostic events and the `orders` table
`id-campus-cafe-func`	Storage Queue Data Contributor	`stcampuscafe`	Send and process `notify` ; write `notify-poison`
Team group	Contributor	`rg-campus-cafe`	Deploy and manage
Team group	Queue / Table Data Contributor + Blob Data Reader	`stcampuscafe`	Inspect messages and rows (Contributor alone gives no data access)

The ACS connection string sits in a Function App setting (encrypted at rest). v1 has no Key Vault.

Telemetry, settings and build order

How the parts talk to each other, what to configure, and the order to build

Telemetry and alerts

Source	Sends	To
Function App	requests, traces, exceptions	Application Insights → `log-campus-cafe`
Storage queue service	StorageRead / Write / Delete (`StorageQueueLogs`)	`log-campus-cafe`
Metric alert	failed requests ≥ 1 in 5 min	Action group → email
Log alert (optional)	PutMessage to `notify-poison`	Action group → email

Queue Message Count is account-wide with hourly grain, so it is not used for alerts.

Key settings

- Host storage by identity: `AzureWebJobsStorage__accountName` , `__credential=managedidentity` , `__clientId` .
- Queue and table connections by identity: `__queueServiceUri` , `__tableServiceUri` (same `__credential` and `__clientId`).
- Queue trigger: 5 tries then `notify-poison` ; `visibilityTimeout` 30–60 s.
- Function App: HTTPS only, max 5 instances, always-ready 0.
- Flex portal wizard: pick **Managed identity** on the Authentication page. It creates the user-assigned identity.

Build order

1. Resource group, Log Analytics, Application Insights, storage account.
2. Function App on Flex (wizard, Managed identity) + role assignments.
3. Table `orders` , queue `notify` .
4. `SubmitOrder` saves to the table. Test in Postman.
5. `SendNotification` with `EMAIL_MODE=log` .
6. `UpdateOrderStatus` with status rules.
7. ACS: Communication Services, Email service, managed domain. `EMAIL_MODE=acs` .
8. Diagnostic setting, alert rules, action group.
9. Budget, then harden (shared-key off, RBAC review).

Scope and limits

- Detail of the v1 data-flow diagram only. No change to design, flow or features.
- Azure-managed email domain: 5 emails/min, 10/hour. ACS Email retires 30 Sep 2028 and Microsoft limits new ACS onboarding from 23 Oct 2026.
- Flex Consumption is Linux only. Pick a region the portal offers.

References and credits

- **Icons:** Microsoft Azure Public Service Icons V24, file `Azure_Public_Service_Icons_V24.zip` (Azure architecture icons, © Microsoft). Used unmodified (no crop, flip, rotate or distortion) with the product name beside each icon, for architecture documentation, as the pack's Terms of Use permit. Icons used: Users, Subscriptions, Resource Groups, App Service Plans, Function Apps, Storage Accounts, Table, Blob Block, Storage Queue, Azure Communication Services, Application Insights, Log Analytics Workspaces, Diagnostics Settings, Managed Identities, Alerts, Cost Budgets.
- **Microsoft documentation:** Resource abbreviations · Flex Consumption how-to · Functions connections and roles · Queue trigger · Queue monitoring reference · ACS service limits · ACS retirement guide